States, assuming 66% resource availability and a 60% conversion efficiency, an additional 1.4 TW (terawatts) of combined solar and wind capacity for the United States alone will be required.

Innovations in renewable project finance will also be required to achieve this scale. For example, high wind and solar-resource geographies that are too far from major cities for cost-effective transmission-line connections will be ideal for fuel generation, particularly to get to TW scale. However, without a transmission line to secondary customers for the power produced, such projects will have to be dedicated to fuel production. Because CO_2-to-fuels production is based on new technologies, technical risk will have to be addressed in such projects. Possible approaches to this could include lines of credit or other such facilities that would be sufficient to pay for transmission lines if needed as insurance against unforeseen reductions in fuel production. Such financial innovations could accelerate these projects, but as with any risk mitigation, they will also have costs. These costs must be kept low enough to be compatible with the competitiveness of the renewable fuels. In order to scale quickly enough to the magnitude necessary for addressing the climate crisis, this challenge must be addressed simultaneously with the challenges of technology scale-up and deployment.

A large number of fuel-making systems will also need to be built and deployed—approximately 150,000 systems rated for one million gallons per year each (MGY) for the United States alone. The existing infrastructure for gasoline, diesel, and jet fuels, however, need not be changed, because renewable CO_2-to-fuel gasoline, diesel, and jet fuel are molecularly identical to their fossil fuel antecedents (although much cleaner, because they do not contain sulfur, benzene, heavy metals, etc.), and could be transported and sold with existing infrastructure. Most importantly, this approach to decarbonizing transportation fuel can happen quickly. This is the kind of solution we need, because we don't have time to waste.

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https://doi.org/10.1016/j.joule.2020.01.002

COMMENTARY

Electric Vehicles Batteries: Requirements and Challenges

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Ted Miller is manager of electrification subsystems and power supply research. His team is responsible for Ford global electrification subsystem and power supply research, delivering battery system design innovations in advanced cell technology, packaging, thermal, EDS, EMC, charging, power conversion, and energy management and modeling. They provide subject matter expertise from raw materials to end-of-life recycling. The team also leads collaboration with university, industrial, and National Lab partners. Mr. Miller is chairman of the United States Advanced Battery Consortium and a member of the Idaho National Laboratory Strategic Advisory Committee and the University of Michigan Energy Institute External Advisory Board.

Introduction

Since the commercialization of lithium-ion batteries (LIBs), tremendous progress has been made to increase energy density, reduce cost, and improve the performance of batteries. The advances in battery technology drive the development of electric vehicles (EVs). As EVs can help eliminate car emissions and reduce petroleum reliance, almost all automakers have launched EV production, and multiple countries/cities have announced plans to ban internal combustion engine (ICE) vehicles in next 10 to 30 years.[1] Global EV sales have grown rapidly in recent years and surpassed two million for the first time in 2018. Nevertheless, the growth of EV adoption is partially driven by government policy and regulation, and the EV market share is still small (~2%). In order to compete with ICE vehicles, EVs still need to overcome some barriers, particularly in battery technology. In this study, we discuss the main requirements and challenges (see the summary in Table 1) to implement batteries in EVs. It is expected that innovation in these areas will address customers' anxieties and enable sustainable growth of EVs.

General Requirements and Challenges of Implementing Batteries in EVs

Energy Density

Driving range is one of the major concerns of customers regarding EVs,[1] and it is mainly determined by the battery energy densities (the amount of energy stored per unit volume or weight). As space and weight in EVs are limited, the batteries with higher energy densities can drive vehicles a longer distance. LIBs have one of the highest energy densities (250–693 Wh/L and 100–265 Wh/kg) of current battery technology, but it is still significantly less that of gasoline. Thus, a large amount of batteries is required to reach 200–300 miles driving range. As the energy densities of LIBs head toward a saturation limit,[2] next-generation batteries (with energy densities >750 Wh/L and >350 Wh/kg) that are beyond LIBs are needed to further increase driving range more effectively. New designs, such as Li-Sulfur, Li-Air, or Mg-ion batteries, have been explored. Although they have higher theoretical energy densities than LIBs have, they suffer from other issues, such as safety (due to dendrite formation) or poor cyclability, which prevent their application in EVs. In addition, battery packs in EVs include not only cells but also other components like busbars, thermal components, and battery management systems. Accounting for this added hardware further reduces the overall pack-level energy densities. Thus, improving both cell design and pack efficiency is critical to increasing energy densities of EV batteries.

Cost

One benefit of EVs is their lower operating costs relative to those of ICE vehicles. Nevertheless, their high upfront purchase price remains one major obstacle for wide EV adoption. As the battery cost determines the EV price, it needs to be reduced to make EVs more affordable. The battery cost has dropped dramatically (currently ~$200/kWh for pack) since the first large-volume EV production. However, to have a comparable cost to that of ICE vehicles, it is generally believed that the battery pack and cell cost should be below $125/kWh and $100/kWh, respectively.[3] The battery pack value stream is extremely complex, consisting of component manufacture, cell production, module production, and pack assembly.[4] Some costs can be reduced by scale or engineering improvement, while a large fraction of cost relies on innovation in areas like raw materials. One way to reduce cost is to use high energy density materials so that a smaller amount of materials is needed. However, as mentioned in the last section, the systems that are beyond LIBs have a higher energy density, but they still need to overcome issues like safety and cyclability prior to their adoption in EVs. Another way is to replace the existing materials with cheaper ones. In current battery chemistry, cobalt is commonly used in cathodes due to its good thermal stability. On the other hand, it is one of the most expensive battery materials since most of it is mined in unstable regions. Eliminating cobalt can

Table 1. Main Requirements and Challenges for EV Batteries

Battery Attributes	Main Requirements	Main Challenges
Energy Densities	>750 Wh/L & >350 Wh/kg for cells	Difficult to find one battery technology that meets all aspects; Tradeoffs must be managed effectively.
Cost	<$100/kWh for cells	
Fast Charge and Power	80% ΔSOC in 15 min	
Life	15 years	
Performance	Minimum impact by environment	
Safety	No fire/flame/rupture/ explosion	

potentially reduce the cost but is not straightforward since battery safety (due to overheating), and cycle life will be impacted without cobalt. Thus, it is challenging to reduce cost while keeping the same performance.

Along with battery materials, the cost associated with manufacturing (e.g., cell production) also needs to be managed to reduce the overall cost of a battery system. For each form factor, there are a lot of process steps in cell production. One of the main challenges is that some steps (such as drying, formation, and aging) take long processing time and affect the yield rate. In addition, it is challenging to accurately measure the composition of active materials and adjust the process parameters correspondingly, which affects the energy density and quality of cells. A smart product line that can inline control materials, process settings and cell quality is needed to improve cell performance and reduce manufacturing cost.

Fast Charging and Power Capability

In addition to driving range, charging time is also central to customer experience. Most current EVs take hours to fully charge. This works fine when charging overnight or charging during work, for example, but it becomes inconvenient when driving a long distance that requires frequent recharging. Recently, EVs have improved the charge capability, which reduces the charging time significantly. However, to reach the goal that EVs charge to 80% ΔSOC (state of charge) within 10 to 15 min, more efforts are required. The main barriers for fast charging are cell chemistry and material properties. In LIBs, lithium ions transport from cathode to anode during charging, and their speed is controlled by intercalation rate and diffusivity of lithium ions. When the charging rate is high, lithium plating occurs at anode surface and cell temperature increases rapidly, which affect battery capacity, life, and safety. Several attempts have been made to increase the charging rate, which include pre-warming up cells to improve electrode kinetics and employing a higher voltage system to reduce the current level. Optimizing charging algorithms that allow fast charge without impact to battery life and safety is needed for future EVs.

Along with fast charging, power capability is another important criterion for EV performance. The power requirement usually depends on vehicle type. For instance, performance-oriented cars and heavy-duty vehicles have different power needs. In some cases, improving power capability has to compromise energy density and increase the cost of thermal/electrical systems, so EV batteries need to balance different aspects of performance.

Lifespan

To maximize battery performance, battery cells in EVs are fully constrained such that they are not only compressed in modules to reduce swelling but also connected to the cooling plate very tightly via thermal interface material to improve cooling efficiency. Replacing a cell from a pack usually requires reassembling the entire module and replacing the damaged cooling plate, which can be expensive. Because of that, EV batteries are generally not serviceable, and their life is expected to be the same as the vehicle life. According to U.S. Advanced Battery Consortium, the target life for EV batteries is 15 years.[3] Here, the battery life generally means the time when its capacity decreases to 80% of its original capacity. Due to strong reactivity between electrolyte and electrode, the capacity loss in batteries, either during cycling or thermally calendaring, is almost inevitable. Common degradation mechanisms include growth of solid electrolyte interface (SEI), lithium plating, lithium ion loss or uneven distribution within electrodes, electrode volume change due to intercalation, and gas formation from side reactions. For each specific cell, its degradation path depends on the composition of electrode and electrolyte as well as on operating conditions. Good understanding of degradation mechanism, appropriate choice of materials, and management of operating conditions help to extend battery life.

As the EV volume increases rapidly in recent years, how to make use of EV batteries after end of life gains a lot of interests. One way is to recycle EV batteries, which is worthwhile when they contain highly valued metals like cobalt, since recycling provides additional sources for supply. Nevertheless, the process of extracting valuable metals from batteries is still currently expensive. To gain scale in recycling, new processes that are cost effective compared with mining need to be developed. Another way is to reuse them in less-demanding applications like stationary energy storage. One challenge of reusing EV batteries is that used EV batteries have dramatically different formats, structure, and chemistries, and the standards to specify their performance are not well defined. Another challenge is that the cost of

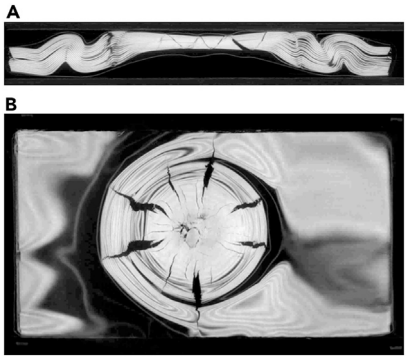


Figure 1. A Pouch Cell with Dimensions of 165 mm, 220 mm, and 5.45 mm in x, y, and z Directions Is Penetrated by a 75 mm Radius Semi-Sphere Quasi-statically

After the penetration reaches ~4.1 mm along the z direction, the X-ray computed tomography scan images in (A) x and (B) z directions show a complex fracture pattern inside the cell.

new batteries keeps decreasing, which makes used batteries less competitive. A combined effort of cell suppliers, automakers, and regulators is required to enable the industry of used batteries.

Performance in Different Environmental Conditions

Since a battery is an electrochemical system that involves reactions and transports of ions and electrons, its performance, such as charging/discharging rate or degradation mechanism, is affected by the environment, particularly temperature. For instance, high temperature allows a relatively high charging rate, but it increases the SEI growth rate, which shortens the battery life. On the other hand, low temperature slows down SEI growth, but it motivates lithium plating. As such, batteries can be optimized for either high or low temperature, but it is difficult to keep the same performance under various temperatures.[4] Since EVs may experience different environments in their lifetime, it is critical to maintain battery performance under different conditions. One way to achieve this goal is to use an effective thermal management system that maintains battery temperature within a certain range. However, it can be a challenge to efficiently cool or heat batteries when needed, due to constraints like limited contact area and heat transfer rate. Optimization of the thermal management design is needed, particularly for batteries of growing energy density and fast-charge capability.

Safety

Safety is one of the most critical factors to EV battery design, since battery failure may lead to a catastrophic consequence. Due to the importance of safety, many countries/regions have regulations for EVs and/or batteries such that they must pass a series of tests (impact, thermal, vibration, etc.) before entering the market.[5] Battery failure can be induced by either external abuse (e.g., crash, penetration, over-charge/discharge, thermal ramp) or internal mechanisms (e.g., lithium dendrite growth). These events may trigger short-circuits or redox reactions at electrodes, which generate heat and activate exothermic reactions, and then can eventually cause thermal runaway.

To mitigate thermal runaway, novel battery materials are developed.[6] However, most of them currently affect cell performance, and tradeoffs should be considered. For instance, aqueous electrolytes are non-flammable but have narrow electrochemical stability window, while solid-state batteries have high mechanical rigidity, but suffer from high interfacial impedance and poor cycle stability. A breakthrough is needed to adopt these materials in EV batteries.

In addition to battery materials, to minimize safety risk, batteries in EVs are equipped with one or more safety features including fuse, vent, current interrupt device, and battery management system that monitors cell voltage and temperature. Moreover, batteries are protected by a structural frame that prevents battery deformation. To optimize the design of a battery system, a large number of tests are usually needed, which can be expensive and time-consuming. On the other hand, computational modeling can predict battery failure in different conditions more efficiently and cost-effectively, which becomes critical in systematic battery design. Nevertheless, battery failure is a complex process (Figure 1), and several challenges remain in battery safety modeling.

The first challenge is to couple multiple physical processes in a single model. A typical battery failure event usually involves different types of physical processes. For instance, an impact process may include material fracture, short-circuit formation, current flow, and thermal propagation. The liquid electrolyte may also affect the mechanical behaviors of jellyrolls. Therefore, the coupled mechanical, electrical, electrochemical, thermal, and fluid responses should be considered in a model. State-of-the-art models[7–9] couple multiple solvers and can capture essential features of battery failure. However, they ignore the influence of electrolyte, and their electrochemical model is not physics-based, which limit their application range.

The second challenge is to measure and model complex material properties accurately. A battery cell generally consists of hundreds of thin layers with significantly different properties. These material properties can be strain-rate and temperature dependent, anisotropic, and affected by electrolyte,[10] which make them a challenge to measure/model. In addition to the stress-strain relationship, to capture battery failure, the failure conditions of each component and short-circuit condition should be known. These critical conditions are difficult to measure/model due to complicated loading and boundary conditions of a cell. Moreover, to get reasonable responses after the onset of short-circuit, short-circuit resistance should be measured, which is not straightforward, since it depends on contact area, stress status, and temperature.

Third, it can be a challenge to scale simulations from component level to pack level. The thickness of individual

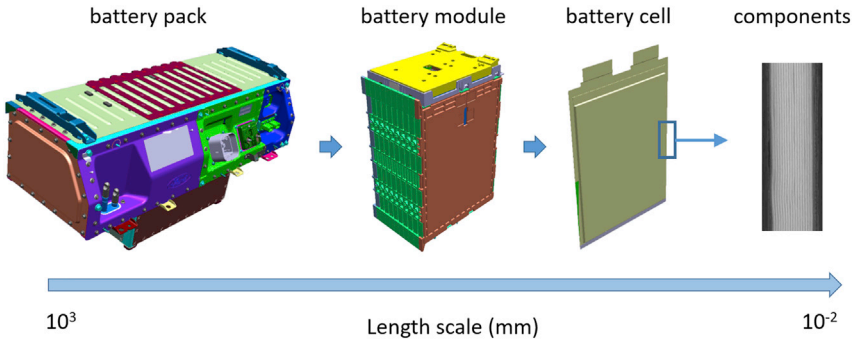


Figure 2. Variation in Length Scale from Battery Pack to Individual Components
The length scale of a battery pack is generally 4 to 5 orders of magitude larger than that of individual layers in a cell.

components in a cell can be several orders of magnitude less than the battery pack size (Figure 2). Therefore, resolving each layer in a module/pack-level simulation is very computationally expensive. Nevertheless, a battery failure event, such as separator fracture or short circuits, usually occurs locally, so the behavior of individual layers should be taken into account in large-scale simulations. Approaches like using a representative sandwich,[9] composite thick-shell elements,[11] or homogenized model[12] can accelerate simulations, but they either ignore the difference in properties of each layer or are not computationally efficient in some solvers, which constrain their capability. Another way to link simulations in different length scales is to utilize sub-modeling. However, this requires appropriate selection of the critical region and transfer of complex boundary conditions between two types of mesh, which can be difficult to set up and validate.

Outlook

In the near future, it will be very challenging for a single battery technology to outperform others in all six aspects. Tradeoffs need to be managed to balance battery performance, life, cost, and safety. As batteries in EVs do not work alone, accurate and efficient system-level analysis is required to optimize the battery system that involves mechanical, thermal, and electrical components. Due to its flexibility and efficiency, computer simulation can play an important role in accelerating systematic battery design. In particular, atomic-scale simulations may help to select key battery materials, while continuum-scale simulation and machine learning may help to reduce design cycles and optimize the system design.

To increase the EV market share substantially, advances in battery technology alone are not sufficient. Other factors like charging infrastructure and government regulations are also critical. Good coverage and capability of charging stations will further reduce customers' concerns, and appropriate government policies can create a healthy environment for the EV market to grow. It is expected that the combined efforts from automakers, battery suppliers, power providers, and governments will help to stimulate demand from customers and accelerate EV adoption.

ACKNOWLEDGMENTS

This work is supported by the Vehicle Technologies Office, Office of Energy Efficiency and Renewable Energy, United States Department of Energy under DOE agreement DE-EE0007288.

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https://doi.org/10.1016/j.joule.2020.01.013